Assignment 5 Task 1 Report

A bidirectional Transformer encoder was implemented from scratch for AG News classification. Six configurations were evaluated. L8_H4 gave the highest Macro F1 of 0.8055; L4_H4 gave the highest accuracy of 80.59%.

Data and experimental setup

Classes: World, Sports, Business and Sci/Tech. Training used 20,000 balanced samples, with 5,000 per class; evaluation used the designated test split (7,600 examples, consistent with saved metrics). Titles and descriptions were joined, lowercased and split on whitespace.

Setting	Value
Vocabulary and sequence	15,000 token limit; [PAD], [UNK], [CLS]; 64 positions
Encoder	d_model = 128; learned positions; FFN width 256; dropout 0.1
Training	Adam; learning rate 0.0001; batch size 64; 10 epochs; seed 42
Studies	L = 2 with h = 2, 4, 8; h = 4 with L = 2, 4, 6, 8

Model and formulas

Token and positional embeddings enter pre layer-normalized attention and feed-forward blocks with residual connections. Padding keys are masked. The final [CLS] state predicts four class logits.

$E = \text{TokenEmbedding}(x) + \text{PositionEmbedding}(\text{position})$

$Q = XW_Q,\ K = XW_K,\ V = XW_V;\ d_k = 128 / h$

$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k} + M_{pad})V$

$\text{MHSA}(X) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W_O$

$\text{Loss} = -(1/N)\ \Sigma_i \log p(y_i \mid x_i)$

$\text{Accuracy} = \text{correct} / N;\ \text{Macro F1} = (1/4)\ \Sigma_c\ 2TP_c / (2TP_c + FP_c + FN_c)$

M_{pad} is 0 for valid keys and approximately $-\infty$ for padding (-10^9 in code). Attention is bidirectional, with no causal mask.

Test results and training curves

Model	Accuracy %	Macro F1	Loss epoch 10	Train time s
L2_H2	80.29	0.8020	0.3940	28.35
L2_H4	79.41	0.7934	0.3427	30.47
L2_H8	80.00	0.7992	0.2886	37.23
L4_H4	80.59	0.8054	0.2493	54.32
L6_H4	79.88	0.7979	0.1710	78.09
L8_H4	80.50	0.8055	0.1429	102.16

More heads did not consistently improve test performance. Two heads gave the best Macro F1 at L = 2. Greater depth lowered training loss, but test improvements were small and nonmonotonic. L8_H4 required about 3.35 times the training time of L2_H4.

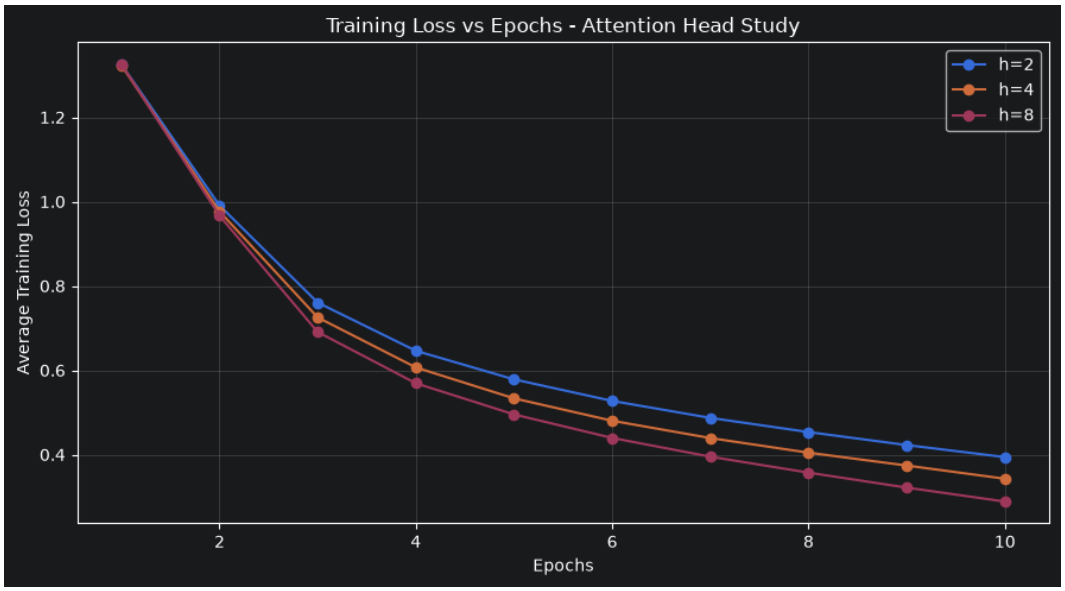


Figure 1. Training loss across attention head counts with two encoder layers.

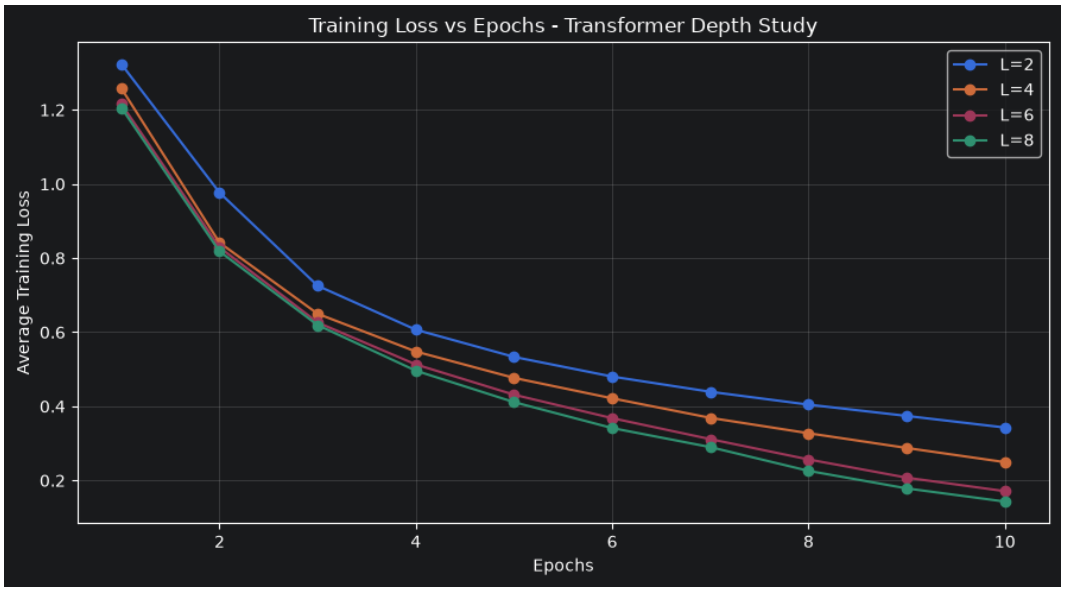


Figure 2. Training loss across encoder depths with four attention heads.

Test confusion matrices and attention

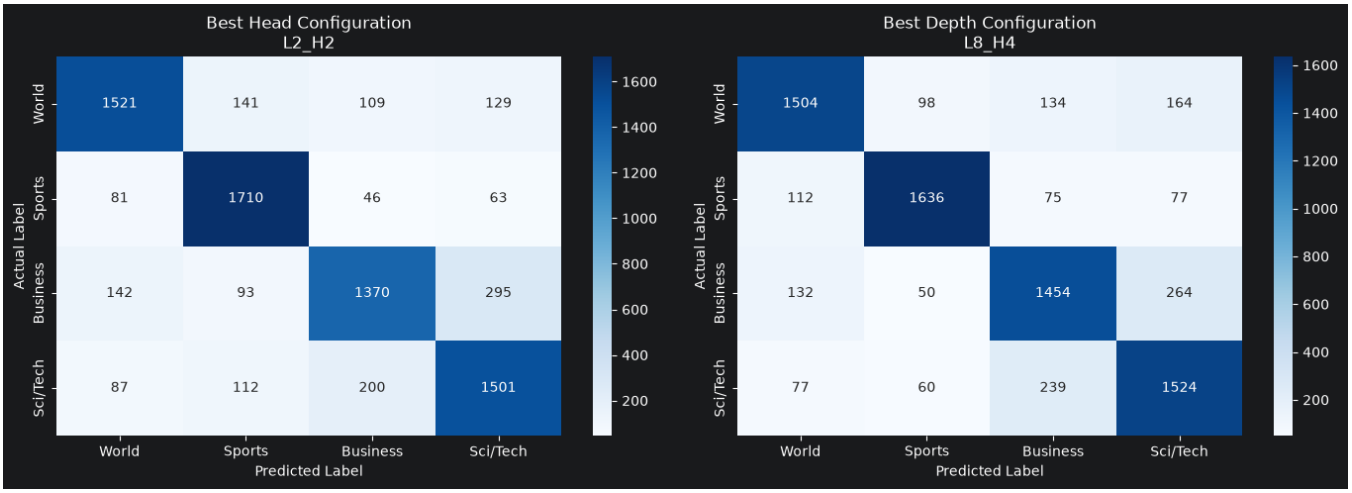


Figure 3. Saved test confusion matrices for the best head configuration L2_H2 and best depth configuration L8_H4, selected by Macro F1. Rows are actual classes; columns are predictions.

The diagonal entries represent correct classifications. Off diagonal entries show errors between categories, allowing class-level inspection beyond overall accuracy.

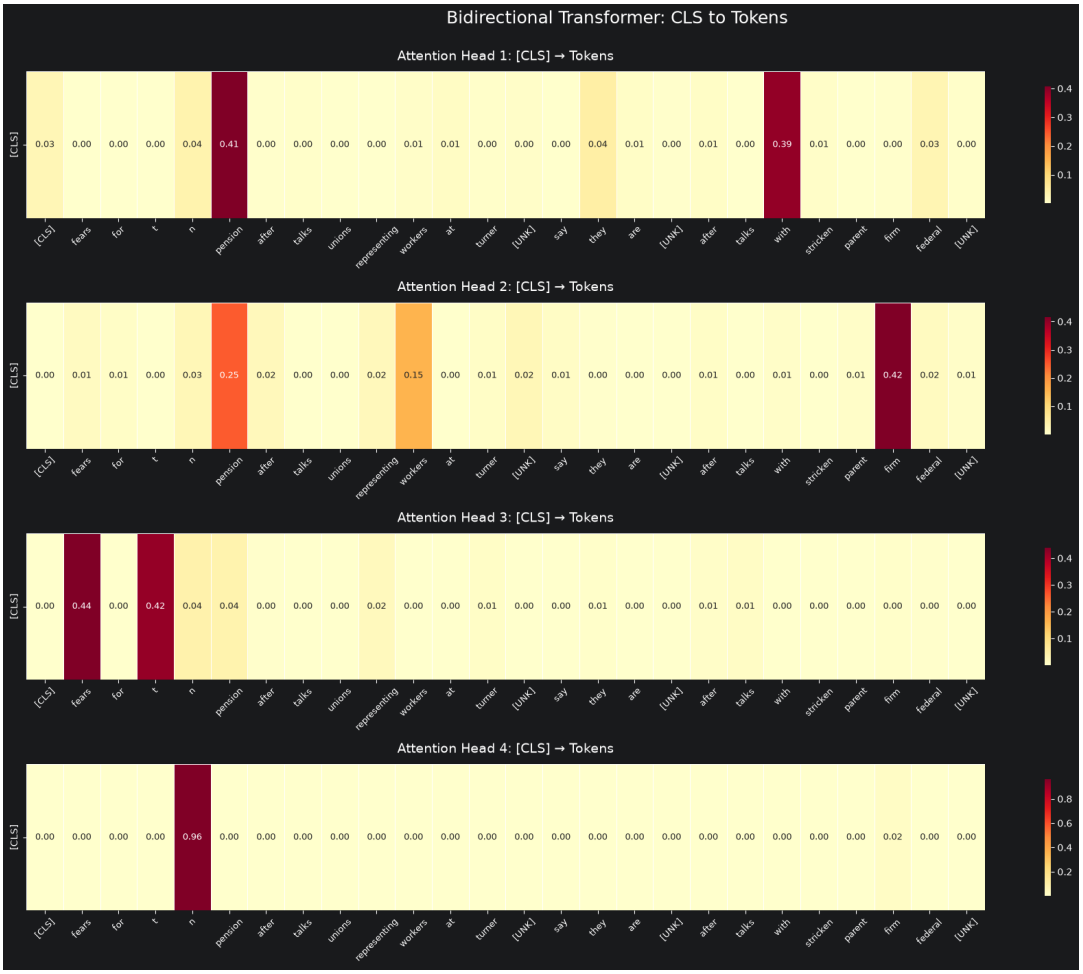


Figure 4. Final layer attention from the [CLS] query to valid tokens in test example 0, across all four heads of L8_H4. Each row is a distribution over token keys.

The heads assign different weights to the same input. The [CLS] representation aggregates token information for classification; padding positions are excluded from the displayed maps.

Attention to the classification token

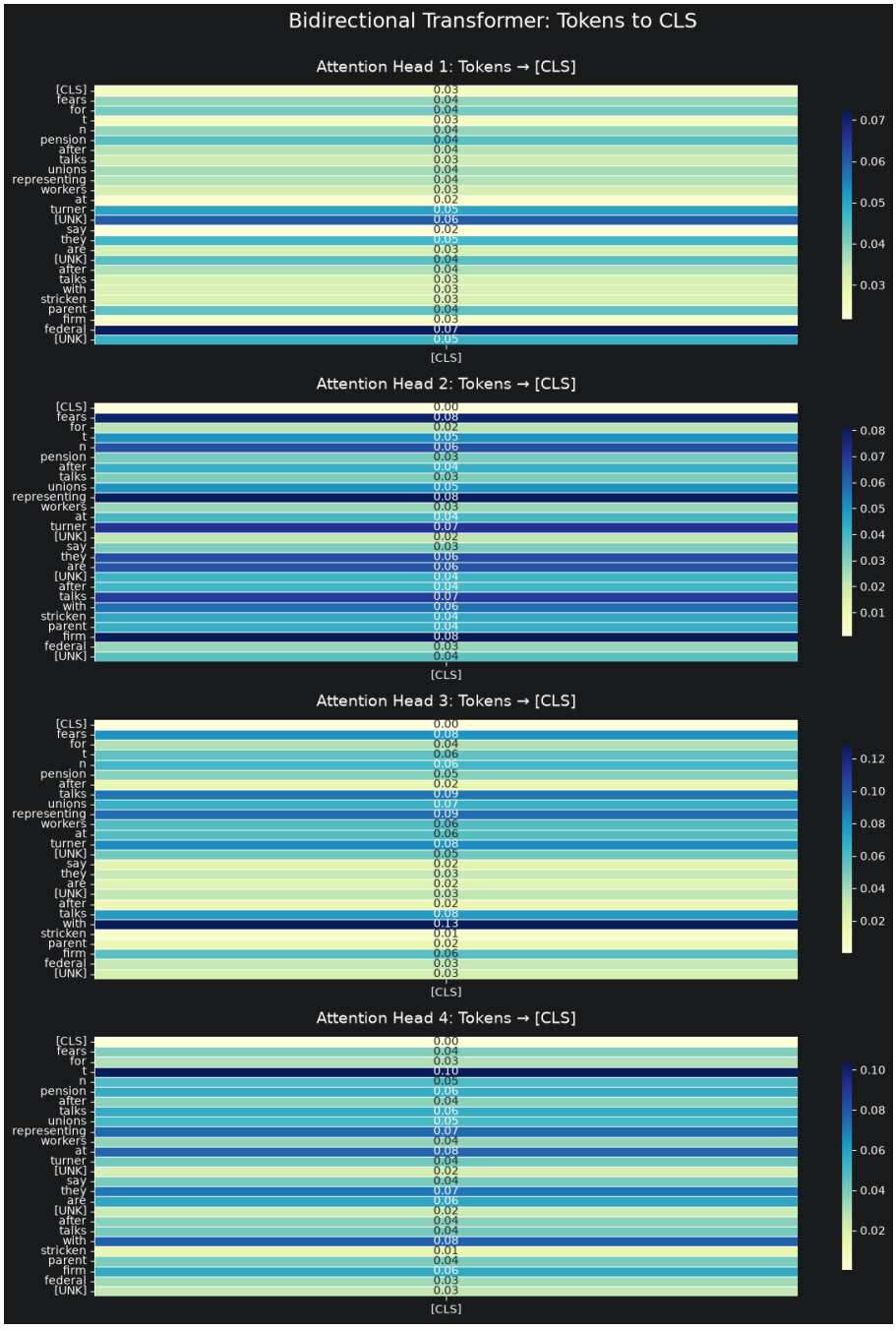


Figure 5. Final layer attention from token queries to the [CLS] key for the same test example and model. Each panel shows one attention head.

Brief observations

L4_H4 offers nearly the same Macro F1 as L8_H4 at about half the training time. Lower training loss alone did not guarantee better test accuracy.

All runs used 10 fixed epochs. The required successive loss threshold of 10^{-4} was not implemented or reached in the printed final epoch losses. No validation curves or repeated seed experiments were saved. This report covers the supplied Task 1 notebook only; Task 2 and positional encoding comparisons are absent.